

NL2TL-Tree: Natural Language to Nested Temporal-Logic Trees for Robustness-Guided Flow-Matching Planning

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Abstract

Letting non-expert users command robots in plain language is a long-standing goal of ubiquitous computing, but free-form instructions are ambiguous and give no guarantee that the resulting behavior is correct. We present **NL2TL-Tree**, an *early-stage prototype* exploring whether a *nested Signal Temporal Logic (STL) tree* can serve as an interpretable, checkable interface between natural-language robot instructions and motion planning. The pipeline is *training-free*—no model is trained or fine-tuned: a frozen LLM is few-shot-prompted to parse a smart-device task into the tree (human-readable and error-localizable to a single sub-node), and a *frozen* graph-encoded flow-matching planner (TeLoGraF), steered by a differentiable STL-robustness term, turns the tree into motion. Test-time mechanisms (operator normalization, nested-tree decomposition, a feasibility self-check that defers to a classical backend) extend the frozen planner, and an online robustness monitor can trigger a stop/re-plan; a *planning-latency-aware* deadline further debits the planner’s own wall-clock from timed goals. We report *preliminary evidence*—a small parsing pilot and a TurtleBot3 Gazebo *simulation*—and outline an evaluation plan for parsing accuracy, failure localization, and user-facing interpretability. Code, prompts, and configs are released.¹

CCS Concepts

• **Computing methodologies** → **Robotic planning**; *Natural language generation*; • **Human-centered computing** → Human computer interaction (HCI).

¹<https://github.com/gjc823878198/NL2TL-Tree-Natural-Language-to-Nested-Temporal-Logic-Trees-for-Robot-Flow-Matching-Planning>

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Keywords

natural language interfaces, signal temporal logic, flow matching, diffusion models, training-free LLM parsing, task planning, verifiable AI, human-robot interaction

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1 Introduction

Pervasive and wearable systems increasingly let people issue tasks in everyday language—“check the kitchen every ten minutes, but never enter the bedroom.” Such instructions are inherently *temporal* and *safety-constrained*, yet most natural-language interfaces translate them with end-to-end models whose output cannot be inspected, checked, or trusted before execution. For ubiquitous computing, where the system acts in a user’s physical space, this lack of interpretability and verifiability is a core barrier to adoption.

Signal Temporal Logic (STL) is an attractive target: it expresses temporal and spatial constraints precisely and admits a quantitative *robustness* of satisfaction. Two threads have grown around it—translating language into formulas [2, 3], and generative planning for STL tasks [7, 9]—but usually in isolation: language work stops at a formula, planning work assumes one is given.

We connect them with one modular pipeline (Fig. 1) built around a single idea—*keep an explicit, verifiable logical structure between the human and the planner*. The entry point is a smart device (e.g., a phone): a user types (or dictates) a task and receives a checked *spatiotemporal command* (where to be, what to avoid, when) that a mobile robot executes. The STL spec enters a frozen flow-matching planner natively through its graph encoder, and a differentiable robustness term steers sampling; nothing is trained, so the train/use gap is closed at test time (§4).

In this poster we present the NL2TL-Tree *prototype*—NL → nested STL tree → robustness monitor → frozen flow-matching planner—built around the nested tree as an *interpretable, checkable* intermediate representation, and report a *preliminary feasibility study* (a frozen-LLM parsing pilot and a Gazebo simulation, §4). The

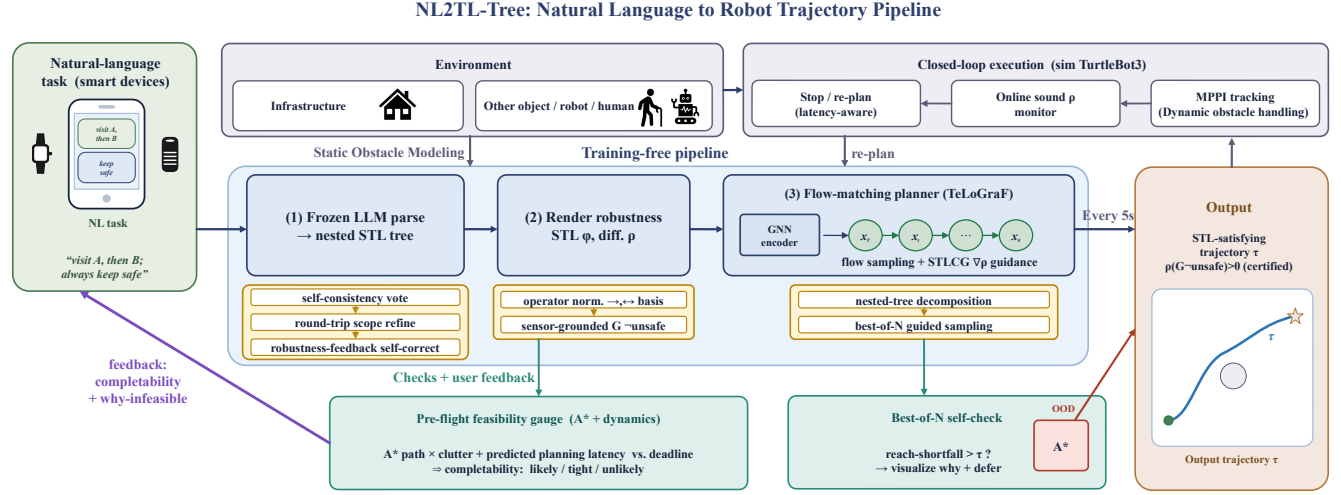


Figure 1: The NL2TL-Tree pipeline—everything frozen / training-free. A smart-device natural-language task is parsed by a frozen LLM into a nested STL tree, rendered to a differentiable robustness ρ , and planned by a frozen flow-matching planner (TeLoGraF) under STL CG robustness-gradient guidance; the certified trajectory is executed closed-loop on a simulated TurtleBot3: the planner models sensed obstacles *statically* while a reactive MPPI tracker avoids the *live*-sensed field (dynamic handling), under an online sound- ρ monitor and a planning-latency-aware stop/re-plan. Each stage is annotated with the test-time mechanisms it uses (§4). A *checks-and-feedback* layer—a pre-flight completeness gauge and a best-of- N self-check that defers out-of-distribution specs to A^* —returns the completeness and a *why-infeasible* explanation to the user.

structural-accuracy levers (self-consistency, round-trip and robustness-feedback self-correction) and a user study are *ongoing*, not yet evaluated.

2 Approach

2.1 Training-free natural language to a nested STL tree

Rather than emitting a flat formula string, we prompt a *frozen* LLM M to produce a *nested logical tree* serialized as JSON (Fig. 2)—atomic predicates at the leaves, Boolean/temporal operators ($\square$, $\diamond$, $\mathcal{U}$, timed variants) at internal nodes. Unlike a flat formula, this representation is *human-readable*, *grammar-checkable* before execution, and *error-localizable* to a single sub-tree—and a round-trip back to an STL string aids verification (Fig. 2, bottom).

By *training-free* we mean *parameter-update-free* (not data-free): no model weights change—the LLM is steered only by K_{shot} in-context exemplars (few-shot, not zero-shot) and the planner stays frozen. The LLM is invoked with a strict-JSON output mode and a prompt assembled from K_{shot} input-output exemplars; we treat K_{shot} as a user-tunable budget ($K_{\text{shot}}=8$ here; larger values trade tokens for coverage of operator diversity). Schema validation (VALIDATESCHEMATIC) rejects malformed JSON or unknown operators before downstream use.

Robustness-feedback self-correction. Rather than accept the first parse, we *verify* it and re-prompt the same frozen LLM when a check fails—a neuro-symbolic loop with no parameter updates. Three verifiers fire cheapest-first: (1) operator arity/schema; (2) grounding consistency (every atom names a known place, no place

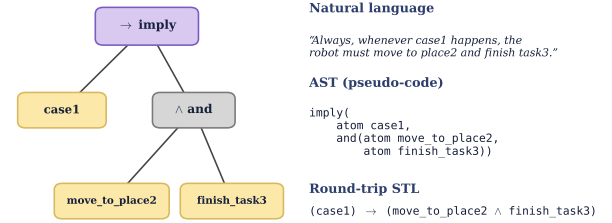


Figure 2: The interpretable intermediate representation. A natural-language instruction (top right) is parsed into a nested STL tree (left); its abstract syntax—serialized as JSON, shown here as compact pseudo-code (right)—is what downstream stages consume, and a round-trip back to an STL string (bottom right) aids verification. The explicit tree exposes operator scope at a glance—the structure the parser must get right and that self-correction repairs.

is both reach and avoid); and (3) *quantitative STL robustness*—we plan a trajectory and, if its robustness ρ cannot be made positive, the spec is unsatisfiable/contradictory and we feed ρ and the offending sub-tree back as a targeted correction. Each failed round appends the rejected JSON plus the diagnostic to the prompt and re-queries (bounded rounds). This directly attacks our dominant error mode (right operators, wrong nesting; see §4); it is implemented, but its quantitative gain is pending the larger study.

Structural-accuracy levers (training-free). Since the residual error is almost purely *structural* (§4), we add two mechanisms that target

the nesting. (i) **Structural self-consistency**: sample K parses at temperature >0 , canonicalize each tree, and majority-vote the structure [13]—operators barely vary across samples while scope does, so the majority recovers nesting that greedy decoding loses to a single near-miss. (ii) **Round-trip scope refinement**: the tree is paraphrased back to NL with its scope made explicit and regenerated only when it disagrees with the instruction [10], turning an invisible inversion into a checkable one.

Role of the public NL–STL dataset. We use a public NL-to-STL dataset three frozen ways: a *few-shot pool* (K_{shot} exemplars); a source of *gold trees* (formulas converted once off-line by a Lark grammar into evaluation-only references); and a *reservoir* of mis-parses written back as future exemplars—none trains a model.

2.2 Rendering to differentiable robustness

The tree compiles to a canonical STL formula φ and a *differentiable robustness* $\rho(\cdot; \varphi)$ under the standard quantitative semantics ($\rho(\neg) = -\rho$, $\rho(\wedge) = \min$, $\rho(\diamond) = \max_t$, dually for $\square, \mathcal{U}$; $\rho > 0 \Leftrightarrow x \models \varphi$; Fig. 3). Following masking-based formulations [8], this smooth ρ is an *optimizable surrogate* that steers sampling; a *separate sound monitor* (exact, non-smoothed) does the validation, with a *duration-aware* keep-safe robustness [6] (worst-case clearance on a satisfied path; penalizing unsafe duration under violation). Optimizer and judge are different functions. We anchor the index axis to wall-clock seconds (arclength at a nominal speed), so $F_{[0,D]} \text{REACH}$ is a real *deadline*; the monitor validates keep-safe and reach-by-deadline on the executed trace. Both margins are in *metres*: ρ_{safe} is worst-case clearance, ρ_{time} the robustness of $F_{[0,D]} \text{REACH}$ within the deadline window (>0 iff the goal is entered before D). Run *online*, a non-positive ρ stops and re-plans (a runtime monitor in simulation), so a violation is repaired, not merely logged.

Planning-latency-aware deadline. The flow-matching planner is not instantaneous (cold-start, then seconds per re-plan), and that wall-clock elapses against $F_{[0,D]}$ —the naive ρ_{time} *freezes* time while the planner thinks. We predict its latency online (cold-start-seeded EMA) and *debit* it: $\rho_{\text{time}}^{\text{lat}} = (D - \hat{t}_{\text{plan}}) - t_{\text{exec}} \leq \rho_{\text{time}}$, handing the tightened deadline to the next re-plan; a plan that “arrives in time” ignoring its own planning cost is then flagged late.

Sensor-grounded keep-safe. Obstacle avoidance is *one* clause $\square \neg \text{UNSAFE}$, with UNSAFE grounded by the union of LiDAR-sensed disks $\{(c_k, r_k)\}$, so $\rho(\square \neg \text{UNSAFE}) = \min_t \min_k (\|p_t - c_k\| - r_k)$ is the worst clearance. A union predicate (vs. N avoid atoms) keeps the GNN encoder in-distribution while the *same* sensed set grounds the keep-safe semantics and drives guidance—so what the robot senses becomes its keep-safe constraint, monitor-validated online ($\rho > 0$) under the sensed obstacle model.

2.3 Tree to trajectory: a frozen flow-matching planner

We pair ρ with a *frozen* generative planner—concretely TeLoGraF [12], which ingests the STL graph through a GNN encoder (the spec enters natively as conditioning) and samples trajectories by flow matching; a differentiable ρ term steers the sampler toward high-satisfaction regions (Fig. 3). The recipe also fits score-based backbones [1, 4, 7, 11]; weights are never updated. **Three test-time**

mechanisms make the frozen planner usable on real specs (Fig. 3), all leaving p_θ untouched. (i) **Operator normalization**: the graph vocabulary lacks $\rightarrow, \leftrightarrow$, so we rewrite them with the standard identities $A \rightarrow B \equiv \neg A \vee B$ and $A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$, verified to preserve robustness *exactly* (zero deviation over thousands of traces and a Boolean truth-table check). (ii) **Nested-tree decomposition**: a nested sequential spec is out-of-distribution, yet each leg is an in-distribution timed reach, so we split the tree, flow-match each leg, and stitch (cf. [5, 9]). (iii) **Feasibility self-check**: from the spread of the best-of- N prior’s reach-shortfall—an *empirical* OOD signal that separates our flow-feasible specs from the out-of-distribution nested one—we defer to a classical planner only when even the best candidate is beyond guidance’s reach (best-of- N per-atom reach-shortfall $> \tau_{\text{OOD}} = 2.5$ m), making the generative-vs-classical choice *measured*, not fixed. The same A^* /dynamics estimate also gives a **pre-flight completability** check—an A^* path through the goals, inflated for clutter and *charged the predicted planning latency*, timed against the deadline—flagging an unreachable deadline up front rather than mid-run.

3 Design Rationale

The pipeline is *designed* for four structural properties, though whether they yield measurable *user* benefits (faster error-spotting, calibrated trust) is a hypothesis we have *not yet user-validated* (§5). It is *interpretable* (the human–planner interface is an explicit nested tree, so a wrong instruction localizes to one sub-node), *checkable* (the sound monitor of §2.2 checks the executed run at the monitor level, $\rho \leq 0$ triggering a stop/re-plan in simulation, with intent aligned by the round-trip check), *training-free* (LLM and planner stay frozen), and *modular* (the typed $\text{NL} \rightarrow \text{tree} \rightarrow \rho \rightarrow \text{trajectory}$ interfaces let any stage be swapped). Unlike NL2TL [3], which fine-tunes a model and stops at a *flat* formula, NL2TL-Tree is training-free, emits a *checkable nested tree*, and carries the spec through to execution; and unlike the planner we reuse—TeLoGraF [12], which assumes the STL is given and is tied to a fixed operator set—we keep it *frozen* and extend it purely at test time.

The user-facing loop. The tree is the interaction surface (Fig. 2): a user types a task on a smart device, sees the parsed nested tree rather than an opaque trajectory, localizes a mis-scoped constraint to one sub-node, edits or re-prompts, and reads a keep-safe / reach-by-deadline *monitor* value ($\rho > 0$; a monitor-level check, not a system-wide guarantee) *before* the robot moves—an *interactive, checkable command interface* rather than a black box. Whether it speeds error-spotting for non-experts is left to a formative study (§5).

4 Case Study and Evaluation

We report a *preliminary* case study (not a full benchmark): a parsing study with a frozen LLM (Stage 1) and an end-to-end planning evaluation in TurtleBot3 *simulation* (Gazebo Classic) (Stage 2).

4.1 Stage 1: NL $\rightarrow$ STL tree (frozen LLM)

Data + protocol. We use the public NL2TL “lifted” dataset (predicates abstracted to $\text{prop}_1, \dots$), covering all eight STL operators and their unbounded variants [3]; each row pairs a sentence with a gold STL formula, which a Lark grammar converts once into a gold tree. We call the frozen LLM in strict-JSON mode (zero training;

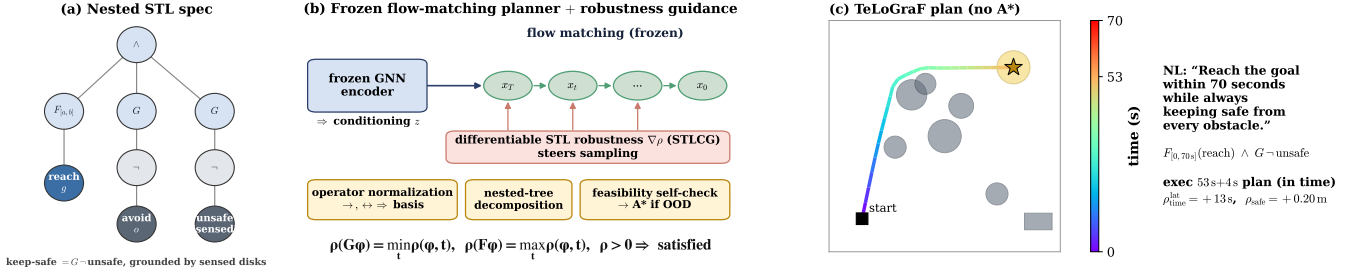


Figure 3: From spec to trajectory (§2.2–2.3). (a) the nested STL tree, with reach (blue) and avoid / keep-safe (grey) atoms; (b) a frozen GNN encoder conditions a flow-matching sampler ($x_T \rightarrow x_0$) that the differentiable STL-robustness gradient $\nabla \rho$ steers, plus the three test-time mechanisms (operator normalization, nested-tree decomposition, feasibility self-check $\rightarrow A^*$); (c) one deployed timed task solved by pure frozen TeLoGraF (no A^*)—the trajectory, coloured by wall-clock seconds, enters the goal at 53 s of a 70 s deadline (latency-aware $\rho_{\text{time}}^{\text{lat}} = +13\text{ s}$, keep-safe $\rho_{\text{safe}} = +0.20\text{ m}$). Per-task results: Table 1.

Table 2); it never sees the formula, and its predicted tree is scored directly against the gold tree. We score the first 30 distinct sentences (no cherry-picking; every parse, including errors, is counted).

Metrics. Beyond strict equality: *op-F1* (operator multiset—right pieces?), *path-F1* (root-to-leaf path multiset—right nesting?), and normalized tree-edit distance ($1 - \text{TED} / \max(|T_p|, |T_g|)$).

Result (preliminary pilot, $n=30$, no fine-tuning). As an initial feasibility check, the frozen LLM usually picks the right operators (**op-F1**=0.95) but *mis-nests scope* (**path-F1**=0.64); exact-match is 56.7% (95% Wilson CI [39, 73]—wide, as expected for a pilot) and the average tree is only 1.03 edits from gold ($\text{TED}_{\text{norm}}=0.84$). The error is thus almost entirely *structural*, which motivates the structural self-consistency voting [13] and round-trip scope refinement [10] levers. A small ablation ($n=30$): self-consistency ($K=5$) helps modestly (exact 0.57 $\rightarrow$ 0.60), while round-trip refinement did *not* (it lowered exact-match). These pilot figures are indicative, pending a larger evaluation.

Flat-STL baseline. A flat-STL baseline (the same frozen LLM emits an STL *string*, parsed by the same grammar; same sentences/gold/metrics) is *no more accurate*: across two further open LLMs the tree matches or beats it on op-F1 (0.94–0.96 vs. 0.90–0.92) and path-F1, trailing only slightly on exact-match (0.60–0.63 vs. 0.67, both grammar-canonicalized). The interpretable structure is essentially free.

What the study stresses. These pilot scores matter less than the *test-time machinery* of §2. Parts of it appear in Fig. 1 but are only sketched here—the *why-infeasible* feedback and the best-of- N self-check visualization—and run live in our demo supplement.

4.2 Stage 2: tree $\rightarrow$ trajectory

We evaluate the frozen-TeLoGraF planner two ways, as an illustrative case study rather than a generalization benchmark.

(i) *Per-task study (cluttered scene; Table 1, Fig. 3(c)).* On six *hand-built* tasks spanning the STL templates, the sound monitor reports $\rho(\Box \neg \text{UNSAFE}) > 0$ and goal entry. An *ablation* (3 re-runs/task): pure flow + STLGC solves the four unnested timed-reach tasks (4/6) but not the nested sequential or OOD response; the full pipeline recovers both (6/6), and disabling the A^* repair drops back to 4/6—the *repair is the load-bearing recovery*, decomposition alone leaving the

Table 1: Per-task Stage-2 outcome on the cluttered scene of Fig. 3 (six *hand-built* tasks). On all six the sound monitor reports keep-safe $\rho(\Box \neg \text{UNSAFE}) > 0$ and goal entry; ρ_{safe} is the mean keep-safe margin over 3 re-runs (full pipeline). The engine is selected automatically: “flow” = pure TeLoGraF + STLGC (no A^*), “+decomp.” = per-leg flow, “ A^* ” = classical fall-back. The lone out-of-distribution spec (trigger_response) is routed to A^* —by design, not a silent success.

Task	STL template	Planner	ρ_{safe} (m)
reach_within_T	$F_{[a,b]} g \wedge \Box \neg u$	flow	+0.20
reach_avoid	$F g \wedge \Box \neg o$	flow	+0.20
reach_goal_north	$F g \wedge \Box \neg o$	flow	+0.20
seq_reach_ABC	$F(A \wedge F(B \wedge F C))$	decomp + A^*	+0.27
cond_reach_either	$\neg F N \rightarrow F S (\equiv F N \vee F S)$	flow	+0.20
trigger_response	$\Box(p \rightarrow F_{[0.5]} q)$	A^* (OOD)	+0.21

count unchanged. Operator normalization separately lets pure flow discharge an implication spec (cond_reach_either). (ii) *Simulated closed loop (TurtleBot3, Gazebo Classic; full setup in Table 2).* On live LiDAR it reaches the goal in 35/53 s ($\rho_{\text{time}} = +0.1$, keep-safe +1.2 m); a *multi-goal* variant visits three regions in order through a cylinder field, all in time.

Metrics. Beyond satisfaction and robustness ρ , we report *inference time*: each re-plan completes in 3.6–4.3 s (mean 3.9, $n=6$) on a laptop CPU, the very latency our planning-aware $\rho_{\text{time}}^{\text{lat}}$ debits (§2.2). Failures are detected, not hidden: the post-hoc check routes a failing plan to A^* repair.

Planned baselines. A study (*not yet run*) will compare against classical LLM $\rightarrow$ STL $\rightarrow$ solver [2], generative [1, 7], and STL/LTL-aware [4, 5, 9, 11, 12] planners.

5 Discussion and Future Work

The structured, training-free middle layer is a promising abstraction for NL robot control: the nested tree is *checkable* and error-localizable to one sub-node, at no parsing-accuracy cost in our pilot. Whether it helps *people* (faster error-spotting, calibrated trust) is the open question our design targets but has not yet user-validated. **Future work** spans a larger parsing benchmark, a Stage-2 baseline study, a formative user study, real-LiDAR hardware, and a better-matched TL robustness.

A Reproducibility

The released code, prompts, and configs (link in the abstract) reproduce the study; Table 2 lists the settings used throughout.

Table 2: Reproducibility settings (frozen LLM + frozen planner).

<i>Parsing (frozen LLM)</i>	
LLM	Llama-3.3-70b-versatile (Groq), strict-JSON
Decoding	greedy (temperature 0), $K_{\text{shot}}=8$
Self-consistency	K samples at temperature 0.7, majority vote
<i>Planning (frozen TeLoGraF)</i>	
Checkpoint	simple_gnn_F (frozen)
Sampling	best-of- N ($N \in [16, 64]$; 8 closed-loop), 64 steps
Guidance	STLCG softmin/softmax $\nabla \rho$
OOD self-check	defer to A* if shortfall $> \tau_{\text{OOD}}=2.5$ m
<i>Closed loop</i>	
Re-plan	every 5 s (MPC-style)
Tracking	MPPI ~ 7 Hz, $v_{\text{max}}=0.22$ m/s
Monitor	stops / re-plans on $\rho \leq 0$ (monitor-level)
<i>Scene</i>	
Goals	reach disks $r \approx 0.45$ m
Obstacles	LiDAR-sensed cylinders $r=0.22$ m

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